Sai Sankar Bhuvanapalli

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Education

Florida State University Tallahassee, FL
Masters of Science in Computer Science *Aug 2023 – May 2025*

- **Relevant Coursework:** Advanced Algorithms, Advanced Operating Systems, Machine Learning, Advanced Data Science, Analytic Methods

Jawaharlal Nehru Technological University Hyderabad, India
Bachelor of Technology in Computer Science & Engineering *Aug 2017 – Jul 2021*

Technical Skills

Programming Languages: Python, Java, SQL, JavaScript
Frameworks & Tools: Spring Boot, AWS Glue, AWS Lambda, AWS Step Functions, **Airflow**, OpenSearch (ElasticSearch), Apache Kafka, **pySpark**, SentenceTransformers, **LangGraph**
Cloud Services: AWS (S3, RDS, DynamoDB, CloudWatch, EC2), PostgreSQL, MySQL, NoSQL, **Gemini API**, Pinecone
Developer Tools: Microservices Architecture, Docker, Kubernetes, **Linux**, Git, Jenkins, Agile Methodologies
Visualization: Tableau, Power BI

Experience

Florida State University ITS Tallahassee, FL
Data Intern *May 2024 – Dec 2024*

- Designed and implemented robust automated data processing pipelines utilizing **Python, SQL, and AWS Cloud Services (Lambda, S3, DynamoDB)**, effectively handling huge datasets with an accuracy exceeding **99%**.
- Integrated with the university's **AIM system** using optimized APIs and SQL improvements, applying **star schema** design to enhance data warehousing and cut manual tasks by **20%**.
- Created and executed comprehensive data validation scripts, establishing automated and scheduled integrity checks using **AWS CloudWatch**, maintaining strict adherence to security protocols and ensuring consistent high-quality data governance.

Infosys - Florida Power & Light Hyderabad, India
Data Engineer *July 2021 – Aug 2023*

- Designed and developed a data pipeline using **Java** and **Spring Boot**, implementing data sorting and routing mechanisms to direct data to analytics platforms and archive unused data in **AWS S3**, reducing overhead by **30%**.
- Engineered **AWS Lambda** functions to process over **1M queries monthly**, reducing query time by **25%**, and built monitoring pipelines that reduced system downtime by **40%**.
- Implemented scalable **NoSQL** storage solutions using **DynamoDB**, and migrated legacy **FTP** pipelines to secure **SFTP** protocols.
- Deployed solutions using **AWS CloudFormation** and **Jenkins CI/CD** to automate provisioning and streamline delivery in containerized environments.

Projects

RAG Roadmap Generator — GitHub

- Built a complete **RAG pipeline** using **Pinecone** for vector search and **SentenceTransformers** for embedding generation.
- Integrated **Gemini API** to synthesize retrieved context into personalized, day-by-day educational roadmaps.
- Implemented scalable batch workflows with robust error handling to support reliable, API-driven processing.
- Used **LangGraph** to deploy and optimize for **asynchronous** and **concurrent processing**, improving system throughput.

Efficient Real-Time Analysis of Reddit Trends— GitHub

- Designed a real-time data pipeline using **AWS Glue** to process API data and generate actionable insights.
- Implemented efficient keyword tracking using the **Count-Min Sketch algorithm**, visualizing insights through interactive dashboards.
- Validated pipeline performance and data accuracy through automated testing and data-quality checks.

Real-Time Notifications for Anomaly Detection— GitHub

- Developed a real-time anomaly alerting system leveraging **AWS SNS** for immediate notifications.
- Designed a responsive UI using **ReactJS** and **TypeScript**, reducing issue resolution time by **30%**.
- Incorporated stakeholder feedback to refine system requirements and efficiently resolve technical issues.